

HW03 1.3~1.7

1. 证明: $\begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} = \begin{bmatrix} A_1 A_2 + B_1 C_2 & A_1 B_2 + B_1 D_2 \\ C_1 A_2 + D_1 C_2 & C_1 B_2 + D_1 D_2 \end{bmatrix}$

$(r+s) \times (p+q) \quad (p+q) \times (m+n)$
 $A_1 = (a_{ij})_{r \times p} \quad B_1 = (b_{ij})_{r \times q} \quad C_1 = (c_{ij})_{s \times p} \quad D_1 = (d_{ij})_{s \times q}$

$A_2 = (a'_{ij})_{p \times m} \quad B_2 = (b'_{ij})_{p \times n} \quad C_2 = (c'_{ij})_{q \times m} \quad D_2 = (d'_{ij})_{q \times n}$

左式 = $\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1p} & b_{11} & \dots & b_{1q} \\ a_{21} & & & & & & \\ \vdots & & & & & & \\ a_{r1} & \dots & a_{rp} & b_{r1} & \dots & b_{rq} \\ c_{11} & \dots & c_{1p} & d_{11} & \dots & d_{1q} \\ \vdots & & & & & \\ \vdots & & & & & \\ c_{s1} & \dots & c_{sp} & d_{s1} & \dots & d_{sq} \end{bmatrix} \begin{bmatrix} a'_{11} & \dots & a'_{1m} & b'_{11} & \dots & b'_{1n} \\ \vdots & & \vdots & \vdots & & \vdots \\ \vdots & & \vdots & \vdots & & \vdots \\ a'_{p1} & \dots & a'_{pm} & b'_{p1} & \dots & b'_{pn} \\ c'_{11} & \dots & c'_{1m} & d'_{11} & \dots & d'_{1n} \\ \vdots & & \vdots & \vdots & & \vdots \\ \vdots & & \vdots & \vdots & & \vdots \\ c'_{q1} & \dots & c'_{qm} & d'_{q1} & \dots & d'_{qn} \end{bmatrix}$

= $\begin{bmatrix} a_{11}a'_{11} + a_{12}a'_{21} + \dots + b_{1q}c'_{q1} & a_{11}a'_{12} + a_{12}a'_{22} + \dots + b_{1q}c'_{q2} & \dots \\ \vdots & \vdots & \\ \vdots & \vdots & \\ c_{s1}a'_{11} + c_{s2}a'_{21} + \dots + d_{sq}c'_{q1} & \dots & c_{s1}b'_{1n} + \dots + d_{sq}d'_{qn} \end{bmatrix}$

右式 = $\begin{bmatrix} \begin{bmatrix} a_{11} & \dots & a_{1p} \\ \vdots & & \vdots \\ a_{r1} & \dots & a_{rp} \end{bmatrix} \begin{bmatrix} a'_{11} & \dots & a'_{1m} \\ \vdots & & \vdots \\ a'_{p1} & \dots & a'_{pm} \end{bmatrix} + \begin{bmatrix} b_{11} & \dots & b_{1q} \\ \vdots & & \vdots \\ b_{r1} & \dots & b_{rq} \end{bmatrix} \begin{bmatrix} c'_{11} & \dots & c'_{1m} \\ \vdots & & \vdots \\ c'_{q1} & \dots & c'_{qm} \end{bmatrix} & \dots & \end{bmatrix}$

= $\begin{bmatrix} a_{11}a'_{11} + a_{12}a'_{21} + \dots + b_{1q}c'_{q1} & a_{11}a'_{12} + a_{12}a'_{22} + \dots + b_{1q}c'_{q2} & \dots \\ \vdots & \vdots & \\ \vdots & \vdots & \\ c_{s1}a'_{11} + c_{s2}a'_{21} + \dots + d_{sq}c'_{q1} & \dots & c_{s1}b'_{1n} + \dots + d_{sq}d'_{qn} \end{bmatrix}$

= 左式 得证

2. 设 a, b, c, d 是实数, 证明矩阵 $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ 可逆当且仅当 $ad \neq bc$,
 当 $ad \neq bc$ 时, 求该矩阵的逆

前推后: 当 $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ 可逆:

$\therefore$ 方阵可逆, 当且仅当它满秩.

$\therefore \begin{bmatrix} a \\ c \end{bmatrix} \begin{bmatrix} b \\ d \end{bmatrix}$ 线性无关

$\therefore ad \neq bc$

后推前: 当 $ad \neq bc$ 时,

$\begin{bmatrix} a \\ c \end{bmatrix}$ 与 $\begin{bmatrix} b \\ d \end{bmatrix}$ 线性无关

方阵秩为 2, 满秩, $\therefore$ 可逆

$$\begin{aligned} \left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array} \right] &\rightarrow \left[\begin{array}{cc|cc} a & b & 1 & 0 \\ a & a \cdot \frac{d}{c} & 0 & \frac{a}{c} \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & \frac{ad-bc}{c} & -1 & \frac{a}{c} \end{array} \right] \\ &\rightarrow \left[\begin{array}{cc|cc} \frac{a}{b} & 1 & \frac{1}{b} & 0 \\ 0 & \frac{ad-bc}{c} & -1 & \frac{a}{c} \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} \frac{a}{b} & 1 & \frac{1}{b} & 0 \\ 0 & 1 & \frac{c}{bc-ad} & \frac{a}{ad-bc} \end{array} \right] \\ &\rightarrow \left[\begin{array}{cc|cc} \frac{a}{b} & 0 & \frac{1}{b} + \frac{c}{ad-bc} & \frac{a}{bc-ad} \\ 0 & 1 & \frac{c}{bc-ad} & \frac{a}{ad-bc} \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & \frac{d}{ad-bc} & \frac{b}{bc-ad} \\ 0 & 1 & \frac{c}{bc-ad} & \frac{a}{ad-bc} \end{array} \right] \end{aligned}$$

$\therefore$ 逆为 $\begin{bmatrix} \frac{d}{ad-bc} & \frac{b}{bc-ad} \\ \frac{c}{bc-ad} & \frac{a}{ad-bc} \end{bmatrix}$

3. 设 $A, D - CA^{-1}B$ 都是可逆矩阵, 证明 $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ 可逆, 并求它的逆

对 $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ 作 Gauss 消元法

$$\begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix}$$

$$\begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I & -A^{-1}B \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{bmatrix}$$

$$\therefore \begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} I & -A^{-1}B \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{bmatrix}$$

$$\therefore \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} I & 0 \\ CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I & A^{-1}B \\ 0 & I \end{bmatrix}$$

$\therefore$ 左式可逆 $\therefore \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ 可逆

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} I & -A^{-1}B \\ 0 & I \end{bmatrix} \begin{bmatrix} A^{-1} & 0 \\ 0 & (D - CA^{-1}B)^{-1} \end{bmatrix} \begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix}$$

4. 设 $\vec{u}, \vec{v}$ 满足 $\vec{u}^T \vec{v} \neq 1$, 证明矩阵 $I - \vec{u} \vec{v}^T$ 可逆, 并求它的逆矩阵

$$\text{设 } \vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_p \end{bmatrix} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_p \end{bmatrix}$$

$$\vec{u}^T \vec{v} = u_1 v_1 + u_2 v_2 + \dots + u_p v_p \neq 1$$

$$\vec{u} \vec{v}^T = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_p \end{bmatrix} \begin{bmatrix} v_1 & v_2 & \dots & v_p \end{bmatrix} = \begin{bmatrix} u_1 v_1 & u_1 v_2 & \dots & u_1 v_p \\ u_2 v_1 & u_2 v_2 & \dots & u_2 v_p \\ \vdots & \vdots & \ddots & \vdots \\ u_p v_1 & u_p v_2 & \dots & u_p v_p \end{bmatrix}$$

$$I - \vec{u} \vec{v}^T = \begin{bmatrix} 1 - u_1 v_1 & -u_1 v_2 & \dots & -u_1 v_p \\ -u_2 v_1 & 1 - u_2 v_2 & \dots & -u_2 v_p \\ \vdots & \vdots & \ddots & \vdots \\ -u_p v_1 & -u_p v_2 & \dots & 1 - u_p v_p \end{bmatrix}$$

假设存在矩阵 $I + \alpha \vec{u} \vec{v}^T$, 使 $(I - \vec{u} \vec{v}^T)(I + \alpha \vec{u} \vec{v}^T) = I$

$$\text{即 } I - \vec{u} \vec{v}^T + \alpha \vec{u} \vec{v}^T - \alpha (\vec{u} \vec{v}^T)(\vec{u} \vec{v}^T) = I \quad (\vec{u}^T \vec{v} = \vec{v}^T \vec{u})$$

$$I + (\alpha - 1 - \alpha \vec{u} \vec{v}^T) \vec{u} \vec{v}^T = I$$

$$\alpha - 1 - \alpha \vec{u} \vec{v}^T = \alpha - 1 - \alpha \vec{u}^T \vec{v} = 0 \quad \alpha = \frac{1}{1 - \vec{u}^T \vec{v}}$$

此时, $(I + \alpha \vec{u} \vec{v}^T)(I - \vec{u} \vec{v}^T)$ 也等于 I $\therefore$ 可逆

$$(\vec{u} \vec{v})^{-1} = I + \frac{1}{1 - \vec{u}^T \vec{v}} \vec{u} \vec{v}^T = \frac{I}{1 - \vec{u}^T \vec{v}}$$

5. 设 L_1, L_2 都是下三角矩阵, L_1, L_2 的对角线元素都是 1, U_1, U_2 都是上三角矩阵, 满足 $L_1 U_1 = L_2 U_2$. 证明: $L_1 = L_2, U_1 = U_2$

设 $L_1 = (l_{ij}) \quad L_2 = (l'_{ij})$ 且 $l_{ii} = l'_{ii} = 1$

设 $U_1 = (u_{ij}) \quad U_2 = (u'_{ij})$

$$(L_1 U_1)_{ij} = \sum_{k=1}^n l_{ik} u_{kj}$$

$$(L_2 U_2)_{ij} = \sum_{k=1}^n l'_{ik} u'_{kj}$$

$$\because L_1 U_1 = L_2 U_2 \quad \therefore \sum_{k=1}^n l_{ik} u_{kj} = \sum_{k=1}^n l'_{ik} u'_{kj}$$

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ l_{21} & 1 & 0 & \cdots & 0 \\ l_{31} & l_{32} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & l_{n3} & \cdots & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} & \cdots & u_{1n} \\ & u_{22} & u_{23} & \cdots & u_{2n} \\ & & u_{33} & \cdots & u_{3n} \\ & & & \ddots & \\ & & & & u_{nn} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ l'_{21} & 1 & 0 & \cdots & 0 \\ l'_{31} & l'_{32} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ l'_{n1} & l'_{n2} & l'_{n3} & \cdots & 1 \end{bmatrix} \begin{bmatrix} u'_{11} & u'_{12} & u'_{13} & \cdots & u'_{1n} \\ & u'_{22} & u'_{23} & \cdots & u'_{2n} \\ & & u'_{33} & \cdots & u'_{3n} \\ & & & \ddots & \\ & & & & u'_{nn} \end{bmatrix}$$

$$\begin{bmatrix} u_{11} & u_{12} & u_{13} & \cdots & u_{1n} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} & \cdots & l_{21}u_{1n} + u_{2n} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} + u_{33} & \cdots & l_{31}u_{1n} + l_{32}u_{2n} + u_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ l_{n1}u_{11} & l_{n1}u_{12} + l_{n2}u_{22} & \cdots & l_{n1}u_{1n} + l_{n2}u_{2n} + \cdots + l_{n,n-1}u_{n-1,n} + u_{nn} \end{bmatrix}$$

$$= \begin{bmatrix} u'_{11} & u'_{12} & u'_{13} & \cdots & u'_{1n} \\ l'_{21}u'_{11} & l'_{21}u'_{12} + u'_{22} & l'_{21}u'_{13} + u'_{23} & \cdots & l'_{21}u'_{1n} + u'_{2n} \\ l'_{31}u'_{11} & l'_{31}u'_{12} + l'_{32}u'_{22} + u'_{33} & \cdots & l'_{31}u'_{1n} + l'_{32}u'_{2n} + u'_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ l'_{n1}u'_{11} & l'_{n1}u'_{12} + l'_{n2}u'_{22} & \cdots & l'_{n1}u'_{1n} + l'_{n2}u'_{2n} + \cdots + l'_{n,n-1}u'_{n-1,n} + u'_{nn} \end{bmatrix}$$

第一行对应元素分别相等, 得

$$u_{11} = u'_{11} \quad u_{12} = u'_{12} \quad \cdots \quad u_{1n} = u'_{1n}$$

第二行对应元素分别相等, 得

$$l_{21} = l'_{21} \quad u_{22} = u'_{22} \quad u_{23} = u'_{23} \quad \cdots \quad u_{2n} = u'_{2n}$$

归纳可得 $l_{ij} = l'_{ij} \quad u_{ij} = u'_{ij}$

$$\therefore L_1 = L_2 \quad U_1 = U_2$$

6. 设 Λ_1 是 p 阶上三角方阵, 对角线元素都是 0; Λ_2 是 q 阶上三角矩阵, 对角线元素都不是 0. 证明: 对 $\forall p$ 行 q 列矩阵 C , 都 $\exists$ 唯一的 p 行 q 列矩阵 B , 使 $B\Lambda_2 - \Lambda_1 B = C$.

$$B = \begin{bmatrix} b_{11} & \dots & b_{1q} \\ \vdots & & \vdots \\ b_{p1} & \dots & b_{pq} \end{bmatrix} \quad \Lambda_1 = \begin{bmatrix} 0 & a_{12} & \dots & a_{1p} \\ & \ddots & & \vdots \\ & & \ddots & a_{p-1,p} \\ & & & 0 \end{bmatrix} \quad \Lambda_2 = \begin{bmatrix} c_{11} & a_{12} & \dots & c_{1q} \\ & \ddots & & \vdots \\ & & \ddots & c_{qq} \end{bmatrix}$$

$$B\Lambda_2 = \begin{bmatrix} b_{11}c_{11} & b_{11}c_{12} + b_{12}c_{22} & \dots & b_{11}c_{1q} + b_{1q}c_{qq} \\ b_{21}c_{11} & b_{21}c_{12} + b_{22}c_{22} & \dots & b_{21}c_{1q} + b_{2q}c_{qq} \\ \dots & \dots & \dots & \dots \\ b_{p1}c_{11} & b_{p1}c_{12} + b_{p2}c_{22} & \dots & b_{p1}c_{1q} + \dots + b_{pq}c_{qq} \end{bmatrix}$$

$$\Lambda_1 B = \begin{bmatrix} a_{12}b_{21} + \dots + a_{1p}b_{p1} & a_{12}b_{22} + \dots + a_{1p}b_{p2} & \dots & a_{12}b_{1q} + \dots + a_{1p}b_{pq} \\ \vdots & \vdots & \ddots & \vdots \\ a_{p-1,p}b_{p1} & a_{p-1,p}b_{p2} & \dots & a_{p-1,p}b_{pq} \\ 0 & 0 & \dots & 0 \end{bmatrix}$$

记 $C = [d_{ij}]$

$$\sum_{k=1}^p b_{ik} c_{kj} - \sum_{k=i+1}^p a_{ik} b_{kj} = d_{ij}$$

$\therefore B\Lambda_2 - \Lambda_1 B = C$ 可转化为关于 b_{ij} 的线性方程组

$\therefore \Lambda_2$ 对角线元素非 0, $\therefore$ 将 Λ_1, Λ_2 构造成的矩阵列满秩

$\therefore$ 线性方程组有唯一解

$\therefore$ 对 $\forall C$, 都 $\exists$ 唯一 B 使 $B\Lambda_2 - \Lambda_1 B = C$

7. 设 $P = \begin{bmatrix} X & Y \\ Z & W \end{bmatrix}$

$$\begin{bmatrix} \Lambda_1 & C \\ & \Lambda_2 \end{bmatrix} \begin{bmatrix} X & Y \\ Z & W \end{bmatrix} = \begin{bmatrix} X & Y \\ Z & W \end{bmatrix} \begin{bmatrix} \Lambda_1 & \\ & \Lambda_2 \end{bmatrix}$$

$$\begin{bmatrix} \Lambda_1 X + CZ & \Lambda_1 Y + CW \\ \Lambda_1 Z & \Lambda_2 W \end{bmatrix} = \begin{bmatrix} X\Lambda_1 & Y\Lambda_2 \\ Z\Lambda_1 & W\Lambda_2 \end{bmatrix}$$

$\therefore$ 只需 $P = \begin{bmatrix} I & B \\ 0 & I \end{bmatrix}$ 即可满足

$$8. \det(A) = a_{11} \cdot a_{22} \cdots a_{nn}$$

前推后: A 可逆 则 $\det(A) \neq 0$ 则 $\forall a_{ii} \neq 0$ 即 A 对角元素都非零

后推前: $\because a_{11}, a_{22}, \dots, a_{nn} \neq 0 \therefore$ 行列式 $\det(A) \neq 0 \therefore A$ 可逆

$$\text{设 } A = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

$$[A|I] = \left[\begin{array}{cccc|cccc} a_{11} & 0 & \cdots & 0 & 1 & & & \\ a_{21} & a_{22} & \cdots & 0 & & 1 & & \\ \vdots & \vdots & \ddots & \vdots & & & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{nn} & & & & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|cccc} 1 & 0 & \cdots & 0 & \frac{1}{a_{11}} & & & \\ a_{21} & a_{22} & \cdots & 0 & & 1 & & \\ \vdots & \vdots & \ddots & \vdots & & & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{nn} & & & & 1 \end{array} \right] \xrightarrow{*} \left[\begin{array}{cccc|cccc} 1 & 0 & \cdots & 0 & \frac{1}{a_{11}} & & & \\ \frac{a_{21}}{a_{22}} & 1 & \cdots & 0 & & \frac{1}{a_{22}} & & \\ \vdots & \vdots & \ddots & \vdots & & & \ddots & \\ \frac{a_{n1}}{a_{nn}} & \frac{a_{n2}}{a_{nn}} & \cdots & 1 & & & & \frac{1}{a_{nn}} \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|cccc} 1 & 0 & \cdots & 0 & \frac{1}{a_{11}} & & & \\ 0 & 1 & \cdots & 0 & -\frac{a_{21}}{a_{11}a_{22}} & \frac{1}{a_{22}} & & \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \\ \frac{a_{n1}}{a_{nn}} & \frac{a_{n2}}{a_{nn}} & \cdots & 1 & \vdots & \vdots & & \end{array} \right]$$

经过行变换变为 $[I|A^{-1}]$, $\therefore$ 右侧矩阵左上角始终为 1 未变过 $\therefore A^{-1}$ 是下三角矩阵

同理可证上三角矩阵的逆是上三角矩阵

观察变换过程可知步骤 $*$ 之后的行变换是将左矩阵左下非对角线元素消为 0 的过程, 对对角线元素无影响 $\therefore$ 对右矩阵对角线元素也无影响 $\therefore A^{-1}$ 的对角元是 A 的对角元的倒数.

同理可证上三角矩阵的逆的对角元是它本身的对角元的倒数

8. 证明下(上)三角方阵 A 可逆当且仅当 A 的对角元都非零, 并且 A 的逆矩阵也是下(上)三角方阵, A^{-1} 的对角元是 A 的对角元的倒数.

9. 试把以下 n 阶方阵分解为下三角矩阵和上三角矩阵的乘积, 并求逆矩阵.

$$\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}, \begin{bmatrix} 2 & -1 & & & \\ 1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 2 & -1 \\ & & & 1 & 2 \end{bmatrix}.$$

$$9. (1) L = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ l_{21} & 1 & 0 & \dots & 0 \\ l_{31} & l_{32} & 1 & \dots & 0 \\ \vdots & & & \ddots & \\ l_{n1} & l_{n2} & l_{n3} & \dots & 1 \end{bmatrix} \quad U = \begin{bmatrix} u_{11} & u_{12} & \dots & u_{1n} \\ & u_{22} & & -u_{2n} \\ & & \ddots & \vdots \\ & & & u_{nn} \end{bmatrix}$$

$$LU = \begin{bmatrix} u_{11} & u_{12} & \dots & u_{1n} \\ u_{11}l_{21} & l_{21}u_{12} + u_{22} & \dots & l_{21}u_{1n} \\ u_{11}l_{31} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

各行分别对应, 得

$$u_{11} = \frac{2}{1} \quad u_{12} = -1 \quad u_{13} = 0 \dots$$

$$u_{22} = \frac{3}{2} \quad u_{23} = -1 \quad u_{24} = 0 \dots \quad u_{2n} = 0$$

$$l_{21} = -\frac{1}{2}$$

$$u_{33} = \frac{4}{3} \quad u_{34} = -1 \quad \dots \quad u_{3n} = 0$$

$$l_{31} = 0 \quad l_{32} = -\frac{2}{3}$$

$$u_{nn} = \frac{n+1}{n}$$

$$l_{n1} = 0 \quad l_{n2} = 0 \quad \dots \quad l_{nn} = 0$$

$$= \begin{bmatrix} 1 & \frac{1}{2} \\ & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \\ & 1 \end{bmatrix}$$

$$\therefore L = \begin{bmatrix} 1 & & & & \\ -\frac{1}{2} & 1 & & & \\ 0 & -\frac{2}{3} & 1 & & \\ \vdots & \vdots & \vdots & \ddots & \\ 0 & 0 & \vdots & \vdots & \frac{n+1}{n} \end{bmatrix} \quad U = \begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ & \frac{3}{2} & -1 & & \\ & & \ddots & & \\ & & & \frac{n+1}{n} & \end{bmatrix}$$

$$A = LU, \quad A^{-1} = U^{-1} \cdot L^{-1}$$

$$= \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \dots & \frac{1}{n} \\ & 1 & \frac{2}{3} & \dots & \frac{2}{n} \\ & & \ddots & \ddots & \vdots \\ & & & 1 & \\ & & & & \frac{n}{n+1} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & & & & \\ \frac{1}{3} & \frac{2}{3} & & & \\ \vdots & \vdots & \ddots & & \\ \frac{1}{n} & \frac{n}{n+1} & \dots & 1 & \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & & & & \\ \frac{1}{3} & \frac{2}{3} & & & \\ \vdots & \vdots & \ddots & & \\ \frac{1}{n} & \frac{n}{n+1} & \dots & 1 & \end{bmatrix}$$

8. 证明下(上)三角方阵 A 可逆当且仅当 A 的对角元都非零, 并且 A 的逆矩阵也是下(上)三角方阵, A^{-1} 的对角元是 A 的对角元的倒数.

9. 试把以下 n 阶方阵分解为下三角矩阵和上三角矩阵的乘积, 并求逆矩阵.

$$U_n = \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}, \quad \begin{bmatrix} 2 & -1 & & & \\ 1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 2 & -1 \\ & & & 1 & 2 \end{bmatrix}$$

$$(2) \quad L = \begin{bmatrix} 1 & & & \\ b_1 & 1 & & \\ & b_2 & \ddots & \\ & & \ddots & b_{n-1} & 1 \end{bmatrix} \quad U = \begin{bmatrix} a_1 & -1 & & \\ & a_2 & \ddots & \\ & & \ddots & -1 \\ & & & a_n \end{bmatrix}$$

由 Doolittle 分解法, 得

$$a_{n+1} = \frac{1}{a_n} + 2 \quad a_1 = 2$$

$$b_{n+1} = \frac{1}{b_n + 2}$$

$$\text{解得 } a_n = \frac{(\sqrt{5}-1)(2\sqrt{5}-3)^n + \sqrt{5}+1}{1-(2\sqrt{5}-3)^n}$$

$$b_n = \frac{\sqrt{5}-1-(2\sqrt{5}-3)^n(\sqrt{5}+1)}{1-(2\sqrt{5}-3)^{n+1}}$$

$$\therefore L = \begin{bmatrix} 1 & & & \\ \frac{1}{2} & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & \frac{\sqrt{5}-1+(2\sqrt{5}-3)^n(\sqrt{5}+1)}{1-(2\sqrt{5}-3)^n} \end{bmatrix} \quad U = \begin{bmatrix} 2 & -1 & & \\ & \ddots & \ddots & \\ & & \ddots & -1 \\ & & & \frac{(\sqrt{5}-1)(2\sqrt{5}-3)^n + \sqrt{5}+1}{1-(2\sqrt{5}-3)^n} \end{bmatrix}$$

$$L^{-1} = \begin{bmatrix} 1 & & & \\ -b_1 & 1 & & \\ b_1 b_2 & -b_2 & \ddots & \\ \vdots & & \ddots & b_{n-1} \\ -b_1 b_2 \dots b_{n-1} & & & 1 \end{bmatrix} \quad U^{-1} = \begin{bmatrix} \frac{1}{a_1} & \frac{1}{a_1 a_2} & \dots & \frac{1}{a_1 a_2 \dots a_n} \\ & \frac{1}{a_2} & & \\ & & \ddots & \\ & & & \frac{1}{a_{n-1} a_n} \\ & & & & \frac{1}{a_n} \end{bmatrix}$$

$$B^{-1} = U^{-1} L^{-1}$$

$$= \begin{bmatrix} \frac{1}{a_1} & \frac{1}{a_1 a_2} & \dots & \frac{1}{a_1 a_2 \dots a_n} \\ & \frac{1}{a_2} & & \\ & & \ddots & \\ & & & \frac{1}{a_{n-1} a_n} \\ & & & & \frac{1}{a_n} \end{bmatrix} \begin{bmatrix} 1 & & & \\ -b_1 & 1 & & \\ -b_1 b_2 & -b_2 & \ddots & \\ \vdots & & \ddots & b_{n-1} \\ -b_1 b_2 \dots b_{n-1} & & & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{a_1} - \frac{b_1}{a_1 a_2} & \dots & -\frac{b_1 b_2 \dots b_{n-1}}{a_1 a_2 \dots a_n} & \dots & \frac{1}{a_1 a_2 \dots a_n} \\ & \ddots & & & \\ & & \ddots & & \\ & & & \ddots & \\ -\frac{b_1 b_2}{a_n} & \dots & \dots & \dots & \frac{1}{a_n} \end{bmatrix}$$

10. 称矩阵 B 是矩阵 A 的**转置**, 记为 $B = A^T$, 如果 $b_{ij} = a_{ji}$ ($\forall i, j = 1, 2, \dots, n$). 称 A 是**对称矩阵**, 如果 $A^T = A$. 称 A 是**反对称矩阵**, 如果 $A^T = -A$. 证明:

(a) 如果矩阵 A 可逆, 则 A^T 也是可逆的, 且 $(A^T)^{-1} = (A^{-1})^T$.

(b) 如果矩阵 A 是可逆的(反)对称阵, 则 A^{-1} 也是(反)对称矩阵.

11. 设矩阵 A 满足 $A^3 = 0$. 证明对任意 $\mu \in \mathbb{R}$, $I - \mu A$ 可逆, 并求它的逆.

12. 设 A, B 是 n 阶方阵, $AB = I_n$. 证明 $B = A^{-1}$.

$$10. (a) \because A \text{ 可逆} \therefore AA^{-1} = I \quad A^{-1}A = I$$

$$\text{两边取转置 } (A^{-1})^T A^T = I \quad A^T (A^{-1})^T = I$$

$\therefore \exists$ 一个矩阵 B 使 $A^T B = B A^T = I \therefore A^T$ 可逆. 逆矩阵为 $(A^{-1})^T$, 等于 $(A^T)^{-1}$

$$(b) \text{ 若 } A \text{ 是可逆对称矩阵, 即 } A^T = A \quad A^{-1} \text{ 存在且 } (A^T)^{-1} = (A^{-1})^T = A^{-1}$$

$\therefore A^{-1}$ 也是对称矩阵

若 A 是可逆反对称矩阵, 即 $A^T = -A$. A^{-1} 存在且

$$(A^T)^{-1} = (A^{-1})^T = (-A)^{-1} = -A^{-1}$$

$\therefore A^{-1}$ 是反对称矩阵

$$11. (I - \mu A)(I + \mu A + \mu^2 A^2) = I - \mu^3 A^3 = I$$

$$(I + \mu A + \mu^2 A^2)(I - \mu A) = I - \mu^3 A^3 = I$$

$\therefore I - \mu A$ 可逆, 逆为 $I + \mu A + \mu^2 A^2$

12. 只需证明 $BA = I_n$

$$BAB = B(AB) = B \cdot I_n = B$$

$$BAB = (BA)B$$

$$\text{则 } (BA)B = B \quad \therefore BA = I_n$$

$$\therefore AB = BA = I_n$$

$$\therefore B = A^{-1}$$

$$13. \begin{bmatrix} 2 & -2 & 1 & -1 & 1 & | & 1 \\ 1 & 2 & -1 & 1 & -2 & | & 1 \\ 4 & -10 & 5 & -5 & 7 & | & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -2 & 1 & -1 & 1 & | & 1 \\ 3 & 0 & 0 & 0 & -1 & | & 2 \\ 4 & -10 & 5 & -5 & 7 & | & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 3 & 0 & 0 & 0 & -1 & | & 2 \\ 2 & -2 & 1 & -1 & 1 & | & 1 \\ 4 & -10 & 5 & -5 & 7 & | & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & -\frac{1}{3} & | & \frac{2}{3} \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} & -\frac{5}{6} & | & \frac{1}{6} \\ 0 & 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{cases} x_1 = \frac{1}{3}x_5 + \frac{2}{3} \\ x_2 = \frac{1}{2}x_3 - \frac{1}{2}x_4 + \frac{5}{6}x_5 + \frac{1}{6} \end{cases}$$

$\therefore$ 自由变量为 x_3, x_4, x_5 . 主变量为 x_1, x_2

$$\therefore \text{解为} \begin{bmatrix} \frac{1}{3}x_5 + \frac{2}{3} \\ \frac{1}{2}x_3 - \frac{1}{2}x_4 + \frac{5}{6}x_5 + \frac{1}{6} \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \quad (x_3, x_4, x_5 \in \mathbb{R})$$

13. 求以下线性方程组的全部解:
$$\begin{cases} 2x_1 - 2x_2 + x_3 - x_4 + x_5 = 1, \\ x_1 + 2x_2 - x_3 + x_4 - 2x_5 = 1, \\ 4x_1 - 10x_2 + 5x_3 - 5x_4 + 7x_5 = 1. \end{cases}$$

14. $\vec{x} = \begin{bmatrix} 1 \\ \lambda \end{bmatrix} \quad (\lambda \in \mathbb{R})$

$A: 2 \times 2$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 1 \\ \lambda \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{cases} a_{11} + \lambda a_{12} = 1 \\ a_{21} + \lambda a_{22} = 3 \end{cases} \quad \text{对 } \forall \lambda \in \mathbb{R} \text{ 恒成立}$$

$$\therefore A = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$$

15. 左推右: 当 $A\vec{x} = \vec{b}$ 有解时设 $\vec{x}$ 为一个解 $A\vec{x} = \vec{b}$

若 $\begin{bmatrix} A^T \\ \vec{b}^T \end{bmatrix} \vec{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ 有解, 设 $\vec{y}_0$ 为其中一个解 则 $A^T \vec{y}_0 = 0, \vec{b}^T \vec{y}_0 = 1$

$\therefore A^T \vec{y}_0 = 0 \quad \therefore \vec{y}_0^T A = 0 \quad \therefore \vec{y}_0^T A \vec{x} = 0$

又 $A\vec{x} = \vec{b} \quad \therefore \vec{y}_0^T \vec{b} = 0$ 这与 $\vec{b}^T \vec{y}_0 = 1$ 矛盾 $\therefore$ 假设不成立 即 $\begin{bmatrix} A^T \\ \vec{b}^T \end{bmatrix} \vec{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ 无解

右推左: 若 $\begin{bmatrix} A^T \\ \vec{b}^T \end{bmatrix} \vec{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ 无解. 假设 $A\vec{x} = \vec{b}$ 无解.

$\therefore r(A) < r[A, \vec{b}] \quad \therefore$ 存在非零向量 $\vec{z}$, 使 $A^T \vec{z} = 0$ 且 $\vec{b}^T \vec{z} \neq 0$

令 $\vec{y} = \frac{\vec{z}}{\vec{b}^T \vec{z}}$ 则有 $A^T \vec{y} = 0$ 且 $\vec{b}^T \vec{y} = 1$ 矛盾 假设不成立 $\therefore A\vec{x} = \vec{b}$ 有解

综上, $A\vec{x} = \vec{b}$ 有解当且仅当 $\begin{bmatrix} A^T \\ \vec{b}^T \end{bmatrix} \vec{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ 有解